Patentanwälte ■ Radlkoferstraße 2 ■ 81373 München

European Patent Office
80298 Munich

Tankred Zimmermann ■ Dipl.-Ing.
Ferdinand Stöckeler ■ Dipl.-Ing.
Franz Zinkler ■ Dipl.-Ing.
Markus Schenk ■ Dipl.-Phys.
Günter Hersina ■ Dipl.-Ing.
Markus Burger ■ Dipl.-Ing.
Oliver Schairer ■ Dipl.-Inf. Univ.
Hannes Pfitzner ■ Dipl.-Ing.
Andreas König ■ Dipl.-Ing. M.Sc.
Julian Schlenker ■ Dipl.-Ing.$^{\Delta}$
Florian Fischer ■ B. Eng.$^{\Delta*}$
Alberto Zuccollo ■ PhD$^{\Delta*}$
Senko Yoshida ■ B. Sc.$^{\Delta*}$
Bastian Miller, Dr. ■ M.Sc.$^{\Delta}$
Andreas Creutz ■ M.Sc.$^{\Delta*}$

March 24, 2026

European Patent Application 25182434.8
Applicant: BEIJING BAIDU NETCOM SCIENCE TECHNOLOGY CO., LTD.
Our Ref.: BAI250601PEP / sy

In reply to the Official Communication pursuant to Rules 70 (2) and 70a (2) EPC dated November 10, 2025:

Request

1. It is herewith indicated that the Applicant wishes to proceed further with the above-referenced patent application.

2. To address the deficiencies noted in the Written Opinion accompanying the European Search Report, it is kindly requested to base the further examination of the present application on new claims 1 to 12 which is enclosed herewith.

Telefon/Telephone +49(0)89/790 445 0
Telefax/Facsimile +49(0)89/790 2215
Telefax/Facsimile +49(0)89/749 969 77
mail@schoppe-zimmermann.com
www.schoppe-zimmermann.com

Kanzleianschrift/Office address:
Radlkoferstraße 2, 81373 München, Germany

$^{\Delta}$ nicht Gesellschafter der Partnerschaft
$^{\Delta}$ not a partner of the partnership
* European Patent Attorney only

Partnerschaftsgesellschaft mbB,
eingetragen in das PR unter Nr. PR 974
Amtsgericht München
Ust-Nr./VAT Registration No. DE 270564910

Amendments

New claims 1 to 12 replace original claims. For the primary Examiner's convenience, a marked-up version of the new claims is enclosed, indicating the amendments when compared to originally filed claims. Deletions are crossed out and additions are underlined.

New claims

Original claim 1 has been amended based on original claims 1 and 2, and the disclosure of the originally filed description to form new claim 1. That is, new claim 1 now includes all the features of original claim 2, and the feature "*the onboard perception model is deployed on the vehicle, other vehicles different from the vehicle, or a server*". The amendment to introduce the feature into claim 1 is supported by the disclosure in paragraph [0056] of the originally filed description of the present application. Following these amendments, original claim 2 has been cancelled, and remaining claims have been renumbered and their dependency has been adapted accordingly.

Original claim 7 has been amended in a similar manner to original claim 1.

Original claim 3 has been amended based on original claim 3 and the disclosure of the originally filed description to form new claim 2. That is, new claim 2 now includes features "*the associated-region lane position comprises coordinates of one or more points of a lane line in the associated region, and the associated-lane spatial feature represents a spatial semantic information of the lane line in the associated region*". This amendment is supported by the disclosure in paragraphs [0089] and [0095] of the originally filed description of the present application.

Addition to the above amendment, the expression of original claim 3 an "*associated spatial feature*" has been amended as an "*associated-lane spatial feature*". In the originally filed description, both of an "*associated spatial feature*" (see, e.g., paragraph [0091]) and an "*associated-lane spatial feature*" (see, e.g., paragraph [0094]) are used for the same meaning. To avoid unnecessary misinterpretation or misunderstanding, the technical wording should be unitary used, and, therefore, the expression an "*associated spatial feature*" in original claim 3 (corresponding to new claim 2) has been replaced by the expression an "*associated-lane spatial feature*".

Further, in response to the clarity objection (see item 2.3 of the Written Opinion), original claims 9 and 10 have been cancelled, and remaining claims have been renumbered.

As demonstrated above, all the amendments are disclosed by the specification as originally filed. The present application therefore has not been amended in such a way that it extends beyond the content of the papers as originally filed in the sense of Article 123 (2) EPC.

Clarity

As indicated above, now claim 1 now includes the definition of the objected term, and, therefore, the meaning of the objected term must be clear for the skilled person.

New claim 2 (corresponding to original claim 3) now includes the features "*the associated-region lane position comprises coordinates of one or more points of a lane line in the associated region, and the associated-lane spatial feature represents a spatial semantic information of the lane line in the associated region*". In addition, the expression "*associated spatial feature*" has been amended as "*associated-lane spatial feature*".

The associated-region lane attribute includes the associated-region lane position, and the associated-region lane position includes coordinates of one or more points of the lane line in the associated region, and is not obtained by feature extraction. In contrast, the associated-lane spatial feature represents a spatial semantic information of the lane line in the associated region, and is obtained by performing a spatial feature extraction on the associated-region lane attribute. Whether in terms of definition or obtaining method, "*associated-region lane position*" is different from "*associated-lane spatial feature*". Therefore, the difference between the "*associated-lane special feature*" and the "*associated-region lane position*" must be clear for the skilled person in the art.

In response to the objection due to lack of clarity, original claims 9 and 10 have been cancelled. Therefore, the claims fulfil the requirements of Article 84 in combination with Rule 43 (2) EPC.

As demonstrated above, all the objection due to lack of clarity has been overcome, and, thus, the present application is now in compliance with the requirements of Article 84 EPC.

Patentability

According to the Written Opinion, the Examiner asserts that the subject-matter of original claim 2 does not involve an inventive step in view of the combination of the disclosure of document D1 (KATEIE Z LUO ET AL: "Augmenting Lane Perception and Topology Understanding with Standard Definition Navigation Maps", XP091655002), the disclosure of document D3 (Li Tianyu ET AL: "Graph-based Topology Reasoning for Driving Scenes", XP093319963) and the disclosure of document D5 (ZHOU JIANG ET AL: "P-MapNet: Far-seeing Map Generator Enhanced by both SDMap and HDMap Prios", XP091701294). However, the applicant respectfully disagrees and believes that the subject-matter of new claims 1 and 6, i.e., the claims including the features of original claim 2, is novel and involving an inventive step in view of the cited prior art documents D1, D3 and D5.

As indicated above, original claim 1 has been amended to incorporate all the features of original claim 2 to form new claim 1, original claim 2 has been cancelled, original claim 7 has been amended adaptively to form new claim 6, and remaining claims have been renumbered.

At least the following technical features in new claim 1 are not disclosed or taught by any cited prior art documents:

"*fusing a feature to be detected and an associated-region feature by using the onboard encoder to obtain a to-be-detected region feature in a bird's eye view space, wherein the feature to be detected is determined according to the information to be detected, and the associated-region feature is obtained by performing a feature extraction on the associated-region lane attribute";* and

"processing the associated-region feature and the to-be-detected region feature by using the onboard decoder based on an attention mechanism to obtain the road perception information."

When evaluating original claim 2, the Examiner equates the "*encoder*" of document D1 with the "*onboard encoder*" of new claim 1. The applicant respectfully disagrees with the Examiner's assertion.

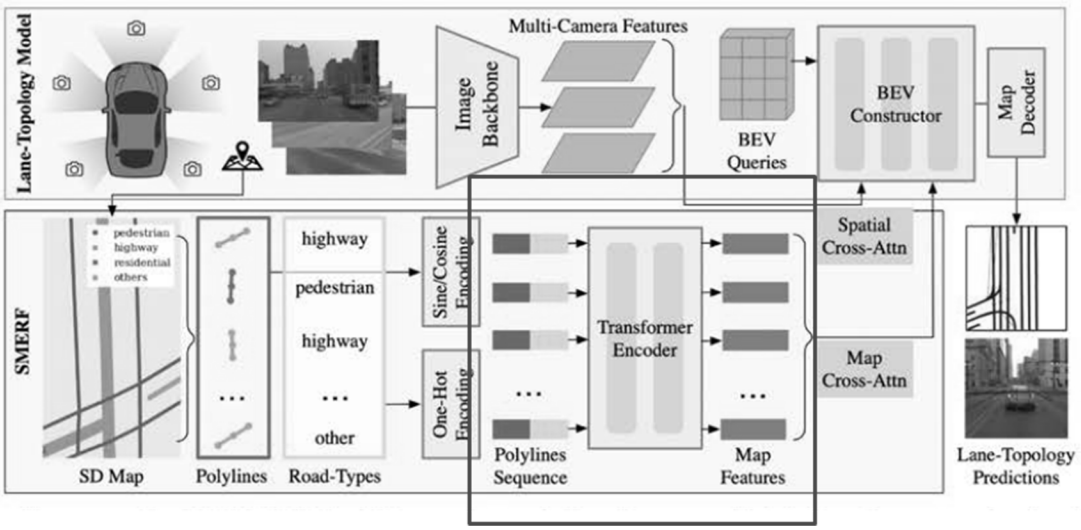


FIG. 2 of document D1

Second paragraph of Section 3 of document D1 recites that "*we first retrieve the SD map, which is encoded into a feature representation using a Transformer encoder*". According to the second paragraph of Section 3 of document D1 and the red box as indicated in FIG. 2 of document D1 (see above), it can be seen that the encoder of document D1 is only used to encode the SD map as a feature representation (i.e. SD map features), and does not fuse the SD map features with any other features, let alone "fuse the SD map features with the features (i.e. the feature to be detected) related to the information to be tested collected by the onboard sensor". On this basis, the encoder of document D1 cannot be used to fuse the SD map feature and the feature to be detected to obtain the to-be-detected region feature in a bird's eye view space.

In contrast, according to the method of new claim 1, a feature to be detected and an associated-region feature are fused by using the onboard encoder to obtain a to-be-detected region feature in a bird's eye view space, wherein the feature to be detected is determined according to the information to be detected collected by the onboard sensor, and the associated-region feature is obtained by performing a feature extraction on the associated-region lane attribute.

In view of above, the operation performed by the "*encoder*" of document D1 is completely different from the operation performed by the "*onboard encoder*" of new claim 1. Therefore,

the "*encoder*" of document D1 cannot be equivalent or corresponding to the "*onboard encoder*" of new claim 1.

Moreover, when evaluating original claim 2, the examiner further equates the "*decoder*" of document D3 with the "*onboard decoder*" of new claim 1. However, the applicant respectfully disagrees with the Examiner's assertion.

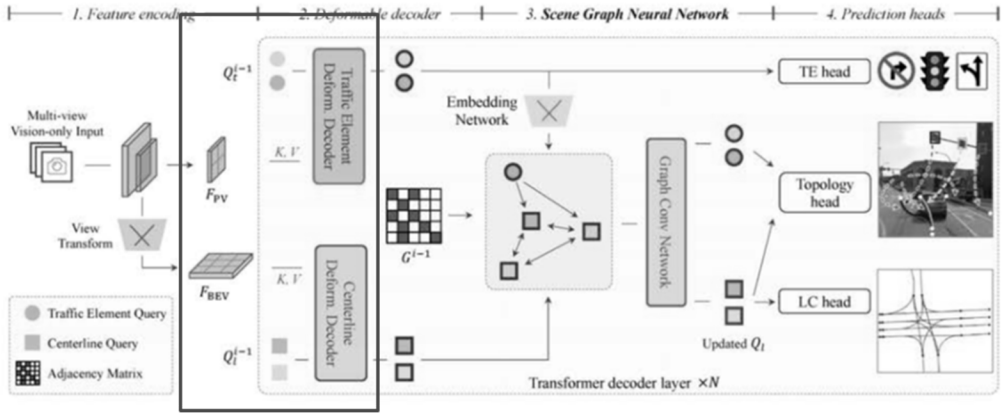


FIG. 2 of Document D3

According to FIG. 2 (as indicated above) and Section 3.2 of document D3 ("*given multi-view images as input, the feature extractor generates multi-scale image features, including the front-view feature F_{PV}, and then converts them into a BEV feature F_{BEV} through a view transformation module; two independent decoders consume F_{PV} and F_{BEV} separately*"), it can be seen that the decoder of document D3 is used to process the front view features and BEV features, where the front view features are obtained by extracting features from the image of the target region, and the BEV features are obtained by converting the front view features using the view conversion module. It can be seen that these two features of document D3 are only related to the target region and not to the associated region, let alone "*the associated-region feature obtained by performing a feature extraction on the associated-region lane attribute*". Therefore, the front view features and BEV features of document D3 cannot be equivalent to the "*associated-region feature*" of new claim 1; Furthermore, the decoder of document D3 is only used to process features related to the target region, rather than features related to the associated region. In addition, in document D3 scheme, the "front view features" and "BEV features" are processed separately using two independent decoders, and the two are not related.

In contrast, according to the method of new claim 1, the associated-region feature and the to-be-detected region feature are processed by using one decoder (i.e. the onboard decoder) based on an attention mechanism to obtain the road perception information, where the associated-region feature is obtained by performing a feature extraction on the associated-region lane attribute.

In view of above, the operation performed by the "*decoder*" of document D3 is completely different from the operation performed by the "*onboard decoder*" of new claim 1. Therefore, the "*decoder*" of document D3 cannot be equivalent or corresponding to the "*onboard decoder*" of new claim 1.

Furthermore, when evaluating original claim 2, the Examiner further equates the "*fusion operation*" of document D5 with the "*fusion operation*" of new claim 1. The applicant respectfully disagrees with the Examiner's assertion.

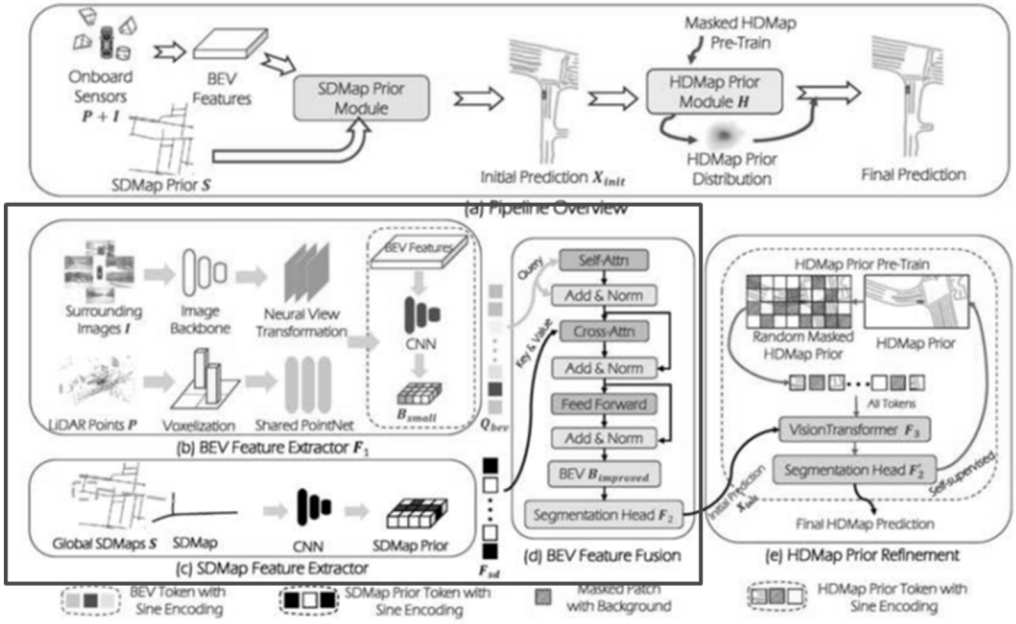


FIG. 2 of document D5

According to FIG. 2 of document D5, as indicated above, it can be seen that the "*fusion operation*" of document D5 is to fuse the BEV feature and the SD map feature; That is to say, the object of document D5's fusion operation is the BEV feature.

In contrast, the "*fusion operation*" of new claim 1 is to fuse the feature to be detected and the associated-region feature to obtain the to-be-detected region feature in the bird's eye view space; That is to say, the result of the fusion operation of new claim 1 is the BEV feature.

In view of above, although both document D5 and new claim 1 involve BEV features in the fusion operation, the BEV feature in document D5 is the object of the fusion operation, while the BEV feature in new claim 1 is the result of the fusion operation. Due to the different roles of BEV features in the fusion operations of document D5 and new claim 1, the "*fusion operation*" of document D5 cannot be equivalent or corresponding to the "*fusion operation*" of new claim 1.

Other cited references D2, D4 and D6 are not argued as relevant prior art documents and they do not disclose or teach the above distinguishing features either, and the Examiner does not deny this either.

In contrast, according to the method of new claim 1, the onboard perception model comprises an onboard encoder and an onboard decoder, and the processing the associated-region lane attribute and the information to be detected by using an onboard perception model to obtain a road perception information of the target region comprises: fusing a feature to be detected and an associated-region feature by using the onboard encoder to obtain a to-be-detected region feature in a bird's eye view space, wherein the feature to be detected is determined according to the information to be detected, and the associated-region feature is obtained by performing a feature extraction on the associated-region lane attribute; and processing the associated-region feature and the to-be-detected region feature by using the onboard decoder based on an attention mechanism to obtain the road perception information.

In this way, the feature to be detected may represent the target region based on an acquisition perspective of the onboard sensor. By fusing the feature to be detected and the associated-region feature using the encoder, it is possible to perform a perspective transformation, based on a preset bird's eye view (BEV) space, on the feature to be detected which is from the perspective of the onboard sensor, and fuse an attribute-semantic information of the feature to be detected and an attribute-semantic information of the associated-region feature. In this manner, the obtained to-be-detected region feature in

the bird's eye view space may represent the region lane attribute of the lane line in the target region from the bird's eye view, so as to avoid errors in the representation of the region lane attribute generated from the image to be detected acquired by the onboard sensor based on the onboard acquisition perspective due to factors such as occlusion and light irradiation, thereby improving the perception accuracy of the subsequent road perception information based on the to-be-detected region feature.

In addition, by processing the associated-region feature and the to-be-detected region feature using the onboard decoder based on the attention mechanism, it is possible to further assist the onboard decoder of the onboard perception model in understanding the semantic information representing the region lane attribute in the to-be-detected region feature based on the associated-region feature, thereby reducing the attribute conflict between the region lane attribute represented by the road perception information and the associated-region lane attribute, and improving the perception accuracy and generation efficiency of the road perception information.

In view of the above, the above distinguishing features of new claim 1 are not disclosed or suggested by all the cited prior art documents alone or in combination. Therefore, new claim 1 involves an inventive step over all the cited prior art documents in any way.

The same arguments also apply to the subject-matter of new claim 6, and, therefore, it is also novel and involving an inventive step over all the cited prior art documents.

Remaining claims directly or indirectly depend on new claim 1 or new claim 6, and, therefore, they are also novel and involving an inventive step in view of the cited prior art documents.

As demonstrated above, none of the cited prior art documents discloses nor suggests the subject-matter of new claims 1 to 12, hence, new claims 1 to 12 must be seen novel and involving an inventive step pursuant to Articles 54 and 56 EPC.

Thus, it is believed that the present case can now proceed to grant. It is, therefore, kindly requested to issue the Communication under Rule 71 (3) EPC.

Closing remarks

The applicant respectfully requests that any amendments to the description be postponed until an agreement has been reached regarding an allowable set of claims.

If only minor amendments are considered to be necessary, the applicant respectfully requests that the Examiner contacts the undersigned attorney by phone so that amended papers may be filed immediately in order to expedite the procedure.

To contact details are as follows:

Telephone: +49 89 790 445 0
e-mail: mail@schoppe-zimmermann.com

Respectfully submitted,

Senko Yoshida

Encl.:
Amended set of claims (annotated and clean versions)